



संख्याएँ

- प्राकृत संख्याएँ : 1, 2, 3, 4, 5, ...
- सम संख्याएँ : 2, 4, 6, 8, 10, ...
- विषम संख्याएँ : 1, 3, 5, 7, 9, ...
- पूर्णांक संख्याएँ : -3, -2, -1, 0, 1, 2, 3, ...
- पूर्ण संख्याएँ : 0, 1, 2, 3, 4, ...
- भाज्य संख्याएँ : 4, 6, 8, 9, ...
- अभाज्य संख्याएँ : 2, 3, 5, 7, 11, ...
- सह-अभाज्य संख्याएँ : (5, 7), (2, 3)
- परिमेय संख्याएँ : $\sqrt{4}$, $\frac{7}{5}$, $\frac{2}{3}$, 3
- अपरिमेय संख्याएँ : $\sqrt{5}$, $\sqrt{7}$, $\sqrt{11}$, $\sqrt{13}$, π
- वास्तविक संख्याएँ : $\sqrt{4}$, $11\frac{4}{7}$
- अवास्तविक संख्याएँ : $\sqrt{-6}$, $\sqrt{-5}$, $\sqrt{-29}$
- सबसे छोटी अभाज्य संख्या है : 2 (दो)
- सबसे छोटी सम अभाज्य संख्या है : 2 (दो)
- सबसे छोटी भाज्य संख्या है : 4 (चार)
- सबसे छोटी पूर्ण संख्या है : 0 (शून्य)
- सबसे छोटी प्राकृत संख्या है : 1 (एक)
- π (पाई) : अपरिमेय संख्या

○ π (पाई) का लगभग मान है : $\frac{22}{7}$ या 3.14159



LEARN THE
CONCEPT IN
60 SECONDS



TRICKBOOK

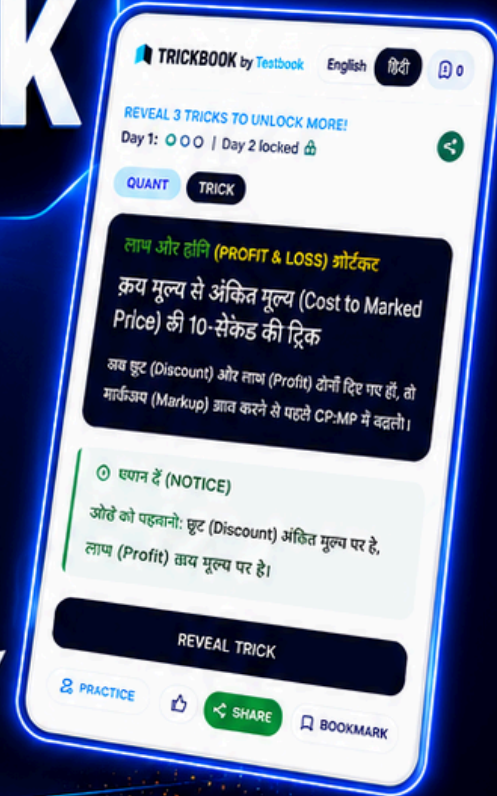
by **Testbook**



AVAILABLE IN
ENGLISH AND **हिंदी**



SAVE YOUR
FAVOURITE TRICK



MORE THAN

10 LAKH+

ASPIRANTS ARE ALREADY
USING THE TRICKBOOK

AAP KAB KAROGE?



CLICK ON THIS TO **VIEW** >>

1. संख्याओं के प्रकार

- प्राकृतिक संख्याएँ (N) : 1, 2, 3, ...
- पूर्ण संख्याएँ (W) : 0, 1, 2, 3, ...
- पूर्णांक (Z) : ... -3, -2, -1, 0, 1, 2, 3 ...
- परिमेय संख्याएँ (Q) : $\frac{p}{q}, q \neq 0$
- अपरिमेय संख्याएँ : $\sqrt{2}, \sqrt{3}, \pi$
- वास्तविक संख्याएँ (R) : परिमेय + अपरिमेय

2. सम और विषम संख्याएँ

$$\begin{aligned}\text{सम} &= 2n \\ \text{विषम} &= 2n + 1\end{aligned}$$

- सम + सम = सम
- विषम + विषम = सम
- सम + विषम = विषम
- विषम × विषम = विषम
- सम × कोई भी = सम



3. अभाज्य और भाज्य संख्याएँ

अभाज्य → ठीक 2 गुणनखंड (1 और स्वयं)
भाज्य → 2 से अधिक गुणनखंड
सबसे छोटी अभाज्य संख्या = 2
एकमात्र सम अभाज्य संख्या = 2

4. गुणनखंड और गुणज

गुणनखंड → पूरी तरह विभाजित करता है
गुणज → संख्या का गुणनफल

(प्र) n के गुणनखंडों की संख्या =

$$\text{यदि } n = p^a q^b r^c$$

$$\text{कुल गुणनखंड} = [(a+1)(b+1)(c+1)]$$

5. म.स. (HCF) और ल.स. (LCM)

म.स. (HCF) → उभयनिष्ठ अभाज्य गुणनखंडों की न्यूनतम घात
ल.स. (LCM) → सभी अभाज्य गुणनखंडों की उच्चतम घात

$$\text{HCF} \times \text{LCM} = \text{दो संख्याओं का गुणनफल}$$

(प्र) यदि HCF = 12, LCM = 360

$$\text{संख्याएँ} = 12x, 12y \Rightarrow (\text{जहाँ } \gcd(x, y) = 1)$$

$$\text{HCF} = 12 \times \frac{360}{12} = 30$$

$$144xy = 4320 \Rightarrow xy = 30$$

$$\text{तब, } xy = 30$$



6. विभाज्यता नियम

- 2 → अंतिम अंक सम
- 3 → अंकों का योग 3 से विभाज्य
- 4 → अंतिम दो अंक 4 से विभाज्य
- 5 → अंतिम अंक 0 या 5
- 6 → 2 और 3 से विभाज्य
- 8 → अंतिम तीन अंक 8 से विभाज्य
- 9 → अंकों का योग 9 से विभाज्य
- 11 → (विषम स्थान का योग - सम स्थान का योग) 11 से विभाज्य

7. शेषफल प्रमेय

यदि n को d से विभाजित करने पर r शेषफल प्राप्त होता है:

$$\Rightarrow n = dq + r$$

(प्र) जब 7^{25} को 6 से विभाजित किया जाए तो शेषफल ज्ञात करें

$$7 \equiv 1 \pmod{6}$$

$$\Rightarrow 7^{25} \equiv 1^{25} \pmod{6}$$

$$\Rightarrow \text{शेषफल} = 1 \text{ (उत्तर)}$$

8. चक्रीयता (अंतिम अंक)

अंतिम अंक 4 के चक्र में दोहराते हैं

$$2 \rightarrow 2, 4, 8, 6$$

$$3 \rightarrow 3, 9, 7, 1$$

$$7 \rightarrow 7, 9, 3, 1$$

(प्र) 7^{101} का अंतिम अंक ज्ञात करें

$$101 \pmod{4} = 1$$

$$\Rightarrow \text{अंतिम अंक} = 7^1 = 7 \text{ (उत्तर)}$$

10. भिन्न

उचित : अंश < हर

अनुचित : अंश ≥ हर

मिश्रित को अनुचित में बदलें

$$\frac{a}{c} = \frac{ac + b}{c}$$

9. म.स. और ल.स. का उपयोग करके संख्या ज्ञात करना

(प्र) दो संख्याओं का म.स. (HCF) = 6 और

ल.स. (LCM) = 180। यदि एक संख्या =

30, तो अन्य संख्या ज्ञात करें।

$$\text{अन्य संख्या} = \frac{6 \times 180}{30} = 6 \times 6 = 36. \text{ (उत्तर)}$$

12. करणी

$$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$$

$$\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$$

11. दशमलव से भिन्न

$$\text{माना } x = 0.333...$$

$$10x = 3.333...$$

$$9x = 3$$

$$x = \frac{3}{9} = \frac{1}{3} \text{ (उत्तर)}$$

13. सरलीकरण युक्तियाँ

- अभाज्य गुणनखंडन का उपयोग करें
- चक्रीयता समय बचाती है
- अंतिम अंक समय बचाता है

13. सरलीकरण युक्तियाँ

- ✓ BODMAS नियम
- ✓ यदि आवश्यक हो तो हमेशा भिन्न में सरल करें

15. सामान्य गलतियाँ

- × पूर्णांकों के चिह्न को अनदेखा करना
- × गलत चक्रीयता
- × HCF और LCM नियमों को मिलाना
- × विभाज्यता में गलत इकाई परिवर्तन

महत्वपूर्ण लघु युक्तियाँ

- ✓ अभाज्य गुणनखंडन का उपयोग करें
- ✓ HCF × LCM ट्रिक
- अंतिम अंक ≠ हमेशा शेषफल



NUMBERS

- ⊙ Natural numbers : 1, 2, 3, 4, 5, ...
- ⊙ Even numbers : 2, 4, 6, 8, 10, ...
- ⊙ Odd numbers : 1, 3, 5, 7, 9, ...
- ⊙ Integer numbers : -3, -2, -1, 0, 1, 2, 3, ...
- ⊙ Whole numbers : 0, 1, 2, 3, 4, ...
- ⊙ Composite numbers : 4, 6, 8, 9, ...
- ⊙ Prime numbers : 2, 3, 5, 7, 11, ...
- ⊙ Co-prime numbers : (5, 7), (2, 3) 
- ⊙ Rational numbers : $\sqrt{4}$, $\frac{7}{5}$, $\frac{2}{3}$, 3
- ⊙ Irrational numbers : $\sqrt{5}$, $\sqrt{7}$, $\sqrt{11}$, $\sqrt{13}$, π
- ⊙ Real numbers : $\sqrt{4}$, $11\frac{4}{7}$
- ⊙ Surreal numbers : $\sqrt{-6}$, $\sqrt{-5}$, $\sqrt{-29}$
- ⊙ The smallest prime number is : 2 (two)
- ⊙ The smallest even prime number is : 2 (two)
- ⊙ The smallest composite number is : 4 (four)
- ⊙ The smallest whole number is : 0 (zero)
- ⊙ The smallest natural number is : 1 (one)
- ⊙ π (pi) : Irrational number 

⊙ π (pi) is approximately : $\frac{22}{7}$ or 3.14159

1. TYPES OF NUMBERS

- Natural Numbers (N) : 1, 2, 3, ...
- Whole Numbers (W) : 0, 1, 2, 3, ...
- Integers (Z) : ... -3, -2, -1, 0, 1, 2, 3 ...
- Rational Numbers (Q) : $\frac{p}{q}$, $q \neq 0$
- Irrational Numbers : $\sqrt{2}$, $\sqrt{3}$, π
- Real Numbers (R) : Rational + Irrational

2. EVEN & ODD NUMBERS

Even = $2n$

Odd = $2n + 1$

- Even + Even = Even
- Odd + Odd = Even
- Even + Odd = Odd
- Odd × Odd = Odd
- Even × Any = Even

**3. PRIME & COMPOSITE NUMBERS**

Prime → Exactly 2 factors (1 & itself)

Composite → More than 2 factors

Smallest prime = 2

Only even prime = 2

4. FACTORS & MULTIPLES

Factor → Divides exactly

Multiple → Product of number

(Q) Number of factors of n =

If $n = p^a q^b r^c$

Total factors = $[(a+1)(b+1)(c+1)]$

5. HCF & LCM

HCF → Lowest power of common primes

LCM → Highest power of all primes

HCF × LCM = Product of two numbers

(Q) If HCF = 12, LCM = 360

Numbers = $12x$, $12y$

HCF = 12 = $12 \times \frac{360}{30}$

LCM = 360 = $12 \times \frac{360}{30}$
= 30

**6. DIVISIBILITY RULES**

- 2 → Last digit even
- 3 → Sum of digits divisible by 3
- 4 → Last two digits divisible by 4
- 5 → Last digit 0 or 5
- 6 → Divisible by 2 & 3
- 8 → Last three digits divisible by 8
- 9 → Sum of digits divisible by 9
- 11 → (Sum of odd place - even place) divisible by 11

7. REMAINDER THEOREMIf n divided by d gives remainder r

$$\Rightarrow n = dq + r$$

(Q) Find remainder when 7^{25} divided by 6

$$7 \equiv 1 \pmod{6}$$

$$\Rightarrow 7^{25} \equiv 1^{25} \pmod{6}$$

$$\Rightarrow \text{Remainder} = 1 \text{ (Ans)}$$

8. CYCLICITY (LAST DIGIT)

Last digits repeat in cycles of 4

2 → 2, 4, 8, 6

3 → 3, 9, 7, 1

7 → 7, 9, 3, 1

(Q) Find last digit of 7^{101}

$$101 \bmod 4 = 1$$

$$\Rightarrow \text{Last digit} = 7^1 = 7 \text{ (Ans)}$$

9. FINDING NUMBER USING HCF & LCM

(Q) Two numbers have HCF = 6

If one number = 30

Find other number

Other = $\frac{6 \times 180}{30} = 6 \times 6 = 36$

11. DECIMAL TO FRACTION

Let $x = 0.333...$

$$10x = 3.333...$$

$$9x = 3$$

$$x = \frac{3}{9} = \frac{1}{3} \text{ (Ans)}$$

13. SIMPLIFICATION TRICKS

- ✓ Use prime factorization
- ✓ Cyclicity saves time
- ✓ Last digit saves time

15. COMMON MISTAKES

- ✗ Ignoring sign of integers
- ✗ Wrong cyclic power
- ✗ Mixing HCF & LCM rules
- ✗ Wrong unit conversion in divisibility

10. FRACTIONS

Proper : numerator < denominator

Improper : numerator $\geq$ denominator

Convert mixed to improper

$$a \frac{b}{c} = \frac{ac + b}{c}$$

12. SURDS

$$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$$

$$\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}} = 5\sqrt{2}$$

13. SIMPLIFICATION TRICKS

- BODMAS rule
- Always simplify to fractions if needed

IMPORTANT SHORTCUTS

- ✓ Use prime factorization
- ✓ HCF × LCM trick
- ✓ Last digit ≠ remainder always

DIVISIBILITY RULES



1 DIVISIBLE BY 1

All numbers are divisible by 1.

Example:

$$18 \div 1 = 18 \quad \checkmark$$

$$100 \div 1 = 100 \quad \checkmark$$



2 DIVISIBLE BY 2

If the last digit is 0, 2, 4, 6, 8 (even number)

Example:

$$526 \quad \checkmark \quad (\text{last digit } 6)$$

$$700 \quad \checkmark \quad (\text{last digit } 0)$$

$$333 \quad \times \quad (\text{last digit } 3)$$

3 DIVISIBLE BY 3

Sum of digits should be divisible by 3

Example:

$$123 \rightarrow 1+2+3 = 6 \quad \checkmark$$

(6 is divisible by 3)

$$512 \rightarrow 5+1+2 = 8 \quad \times$$

(8 is not divisible by 3)

4 DIVISIBLE BY 4

Last two digits should be divisible by 4 or be 00

Example:

$$724 \rightarrow 24 \quad \checkmark$$

$$800 \rightarrow 00 \quad \checkmark$$

$$913 \rightarrow 13 \quad \times$$

5 DIVISIBLE BY 5

Last digit should be 0 or 5

Example:

$$575 \quad \checkmark$$

$$120 \quad \checkmark$$

$$334 \quad \times$$

6 DIVISIBLE BY 6

Number should be divisible by both 2 and 3

Example:

$$108 \quad \checkmark \quad (\text{even, sum} = 9, \text{ 9 is divisible by } 3)$$

$$109 \quad \times \quad (\text{not divisible})$$



7 DIVISIBLE BY 7

Double the last digit and subtract from the remaining number

Example:

$$133 \rightarrow 3 \times 2 = 6; 13 - 6 = 7 \quad \checkmark$$

$$672 \rightarrow 2 \times 2 = 4; 67 - 4 = 63 \quad \checkmark$$

8 DIVISIBLE BY 8

Last three digits should be divisible by 8 or be 000

Example:

$$2832 \rightarrow 832 \quad \checkmark$$

$$5000 \rightarrow 000 \quad \checkmark$$

$$7111 \rightarrow 111 \quad \times$$

9 DIVISIBLE BY 9

Sum of digits should be divisible by 9

Example:

$$729 \rightarrow 7+2+9 = 18 \quad \checkmark$$

$$512 \rightarrow 5+1+2 = 8 \quad \times$$



11 DIVISIBLE BY 11

Difference between sum of alternate digits should be 0 or multiple of 11

Example:

$$1331 \rightarrow (1+3) - (3+1) = 0 \quad \checkmark$$

$$2728 \rightarrow (2+2) - (7+8) = -11 \quad \checkmark$$

12 DIVISIBLE BY 12

Number should be divisible by both 3 and 4

Example:

$$144 \quad \checkmark \quad (\text{divisible by } 3 \text{ and by } 4)$$



13 DIVISIBLE BY 13

Last digit by 4 and remaining number

Example:

$$3 \times 4 = 12;$$

$$14 + 12 = 26 \quad \checkmark$$



17 DIVISIBLE BY 17

Multiply last digit by 5 and subtract from the remaining number

Example:

$$289 \rightarrow 9 \times 5 = 45;$$

$$28 - 45 = -17$$

$$(-17 \text{ is divisible by } 17) \quad \checkmark$$

23 DIVISIBLE BY 23

Multiply last digit by 7 and add to the remaining number

Example:

$$529 \rightarrow 9 \times 7 = 63;$$

$$52 + 63 = 115 \rightarrow$$

$$5 \times 7 = 35; 11 + 35 = 46 \quad \checkmark$$



SQUARE NUMBERS

[1 TO 100]

$1^2 = 1$
$2^2 = 4$
$3^2 = 9$
$4^2 = 16$
$5^2 = 25$
$6^2 = 36$
$7^2 = 49$
$8^2 = 64$
$9^2 = 81$
$10^2 = 100$
$11^2 = 121$
$12^2 = 144$
$13^2 = 169$
$14^2 = 196$
$15^2 = 225$
$16^2 = 256$
$17^2 = 289$
$18^2 = 324$
$19^2 = 361$
$20^2 = 400$
$21^2 = 441$
$22^2 = 484$
$23^2 = 529$
$24^2 = 576$
$25^2 = 625$

$26^2 = 676$
$27^2 = 729$
$28^2 = 784$
$29^2 = 841$
$30^2 = 900$
$31^2 = 961$
$32^2 = 1024$
$33^2 = 1089$
$34^2 = 1156$
$35^2 = 1225$
$36^2 = 1296$
$37^2 = 1369$
$38^2 = 1444$
$39^2 = 1521$
$40^2 = 1600$
$41^2 = 1681$
$42^2 = 1764$
$43^2 = 1849$
$44^2 = 1936$
$45^2 = 2025$
$46^2 = 2116$
$47^2 = 2209$
$48^2 = 2304$
$49^2 = 2401$
$50^2 = 2500$

$51^2 = 2601$
$52^2 = 2704$
$53^2 = 2809$
$54^2 = 2916$
$55^2 = 3025$
$56^2 = 3136$
$57^2 = 3249$
$58^2 = 3364$
$59^2 = 3481$
$60^2 = 3600$
$61^2 = 3721$
$62^2 = 3844$
$63^2 = 3969$
$64^2 = 4096$
$65^2 = 4225$
$66^2 = 4356$
$67^2 = 4489$
$68^2 = 4624$
$69^2 = 4761$
$70^2 = 4900$
$71^2 = 5041$
$72^2 = 5184$
$73^2 = 5329$
$74^2 = 5476$
$75^2 = 5625$

$76^2 = 5776$
$77^2 = 5929$
$78^2 = 6084$
$79^2 = 6241$
$80^2 = 6400$
$81^2 = 6561$
$82^2 = 6724$
$83^2 = 6889$
$84^2 = 7056$
$85^2 = 7225$
$86^2 = 7396$
$87^2 = 7569$
$88^2 = 7744$
$89^2 = 7921$
$90^2 = 8100$
$91^2 = 8281$
$92^2 = 8464$
$93^2 = 8649$
$94^2 = 8836$
$95^2 = 9025$
$96^2 = 9216$
$97^2 = 9409$
$98^2 = 9604$
$99^2 = 9801$
$100^2 = 10000$



TRICK TO REMEMBER

Every square number ends with the square of the number's unit digit.

EXAMPLES:

23^2 ends with $3^2 = 9$
 47^2 ends with $7^2 = 49$
 80^2 ends with $0^2 = 0$

Keep Practicing!



PERCENTAGE CHEAT-SHEET

Formula

$$\text{Percentage} = \left(\frac{\text{Part}}{\text{Whole}} \right) \times 100$$

Column 1

- ① $\frac{1}{3} = 33.33\%$
- ② $\frac{2}{3} = 66.66\%$
- ③ $\frac{1}{4} = 25\%$
- ④ $\frac{2}{4} = 50\%$
- ⑤ $\frac{3}{4} = 75\%$
- ⑥ $\frac{1}{5} = 20\%$
- ⑦ $\frac{2}{5} = 40\%$
- ⑧ $\frac{3}{5} = 60\%$
- ⑨ $\frac{4}{5} = 80\%$
- ⑩ $\frac{1}{6} = 16.66\%$
- ⑪ $\frac{2}{6} = 33.33\%$
- ⑫ $\frac{3}{6} = 50\%$

Column 2

- ① $\frac{5}{6} = 83.33\%$
- ② $\frac{1}{7} = 14.28\%$
- ③ $\frac{2}{7} = 28.56\%$
- ④ $\frac{3}{7} = 42.84\%$
- ⑤ $\frac{4}{7} = 57.14\%$
- ⑥ $\frac{5}{7} = 71.42\%$
- ⑦ $\frac{6}{7} = 85.71\%$
- ⑧ $\frac{1}{8} = 12.5\%$
- ⑨ $\frac{2}{8} = 25\%$
- ⑩ $\frac{3}{8} = 37.50\%$
- ⑪ $\frac{4}{8} = 50\%$
- ⑫ $\frac{5}{8} = 62.50\%$

Column 3

- ① $\frac{7}{8} = 87.50\%$
- ② $\frac{1}{9} = 11.11\%$
- ③ $\frac{2}{9} = 22.22\%$
- ④ $\frac{3}{9} = 33.33\%$
- ⑤ $\frac{4}{9} = 44.44\%$
- ⑥ $\frac{5}{9} = 55.55\%$
- ⑦ $\frac{6}{9} = 66.66\%$
- ⑧ $\frac{7}{9} = 77.77\%$
- ⑨ $\frac{8}{9} = 88.88\%$
- ⑩ $\frac{1}{11} = 9.09\%$
- ⑪ $\frac{2}{11} = 18.18\%$
- ⑫ $\frac{3}{11} = 27.27\%$

Column 4

- ① $\frac{5}{11} = 45.45\%$
- ② $\frac{6}{11} = 54.54\%$
- ③ $\frac{7}{11} = 63.63\%$
- ④ $\frac{8}{11} = 72.72\%$
- ⑤ $\frac{9}{11} = 81.81\%$
- ⑥ $\frac{10}{11} = 90.90\%$
- ⑦ $\frac{1}{12} = 8.33\%$
- ⑧ $\frac{1}{13} = 7.69\%$
- ⑨ $\frac{1}{14} = 7.14\%$
- ⑩ $\frac{1}{15} = 6.66\%$
- ⑪ $\frac{1}{16} = 6.25\%$
- ⑫ $\frac{1}{24} = 4.16\%$

Tip:

Learn Column 1 & 2 well, others become easy!

Remember

- $\frac{1}{2} = 50\%$
- $\frac{1}{10} = 10\%$
- $1\% = \frac{1}{100}$



testbook

Your One-stop
Solution for
Smart Preparation!



testbook

समय, चाल एवं दूरी

आपको आवश्यक एकमात्र सूत्र-पत्र



मूल सूत्र

चाल (Speed) = दूरी ÷ समय

दूरी (Distance) = चाल × समय

समय (Time) = दूरी ÷ चाल



स्मरण रखें:

- जिस राशि का मान ज्ञात करना हो, उसे दक दें।
- शेष दो राशियों से उत्तर प्राप्त करें।

इकाइयाँ (Units):

चाल → किमी/घंटा (km/h) या मी/से (m/s)

दूरी → किमी (km) या मीटर (m)

समय → घंटा (h) या सेकंड (s)

1. इकाई रूपांतरण (Unit Conversion)

$$1 \text{ किमी/घंटा} = \frac{5}{18} \text{ मी/से}$$

$$1 \text{ मी/से} = \frac{18}{5} \text{ किमी/घंटा}$$

2. सापेक्ष चाल (Relative Speed)

• समान दिशा में: सापेक्ष चाल = $S_1 - S_2$

• विपरीत दिशा में: सापेक्ष चाल = $S_1 + S_2$

3. मिलन बिंदु अवधारणा (Meeting Point Concept)

(i) एक-दूसरे की ओर गति



$$\text{समय} = \frac{\text{दूरी}}{S_1 + S_2}$$

(ii) समान दिशा में गति (जहाँ $S_1 > S_2$)



$$\text{समय} = \frac{\text{दूरी}}{S_1 - S_2}$$

उदाहरण:

दो व्यक्ति A एवं B से 100 किमी की दूरी पर चलना शुरू करते हैं। उनकी चाल 40 एवं 60 किमी/घंटा है।

$$\text{समय} = \frac{100}{40 + 60} = \frac{100}{100} = 1 \text{ घंटा}$$

4. स्थिर दूरी अवधारणा (Distance Constant Concept)

यदि दूरी स्थिर हो, तो:

$$S_1 T_1 = S_2 T_2$$

उदाहरण:

यदि चाल का अनुपात 4 : 3 हो, तो समय का अनुपात 3 : 4 होगा।

5. स्थिर चाल अवधारणा (Speed Constant Concept)

यदि चाल स्थिर हो, तो:

$$\frac{D_1}{D_2} = \frac{T_1}{T_2}$$

6. औसत चाल (Average Speed)

$$\text{औसत चाल} = \frac{\text{कुल दूरी}}{\text{कुल समय}}$$

यदि समान दूरी दो अलग-अलग चालों a तथा b से तय की जाए:

$$\text{औसत चाल} = \frac{2ab}{a + b}$$

7. रेलगाड़ी पर आधारित प्रश्न (Problems on Trains)

(i) खम्भा / व्यक्ति को पार करना



$$\text{समय} = \frac{\text{रेलगाड़ी की लंबाई}}{\text{चाल}}$$

(ii) प्लेटफॉर्म को पार करना



$$\text{समय} = \frac{\text{रेलगाड़ी की लंबाई} + \text{प्लेटफॉर्म की लंबाई}}{\text{चाल}}$$

(iii) विपरीत दिशा में दो रेलगाड़ियाँ



$$\text{समय} = \frac{L_1 + L_2}{S_1 + S_2}$$

(iv) समान दिशा में दो रेलगाड़ियाँ (जहाँ $L_1 > L_2$ तथा $S_1 > S_2$)



$$\text{समय} = \frac{L_1 - L_2}{S_1 - S_2}$$

उदाहरण:

120 मीटर लंबी रेलगाड़ी किसी व्यक्ति को 8 सेकंड में पार करती है।

$$\text{चाल} = \frac{120}{8} = 15 \text{ मी/से} = 54 \text{ किमी/घंटा}$$

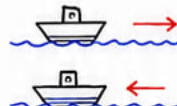
8. नाव एवं धारा (Boats and Streams)

स्थिर जल में नाव की चाल = B

धारा की चाल = S

अनुप्रवाह (Downstream) = $B + S$

प्रतिप्रवाह (Upstream) = $B - S$



$$\text{नाव की चाल} = \frac{D + U}{2}, \text{ धारा की चाल} = \frac{D - U}{2}$$

जहाँ, D = अनुप्रवाह चाल, U = प्रतिप्रवाह चाल

उदाहरण:

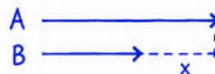
अनुप्रवाह = 15 किमी/घंटा, प्रतिप्रवाह = 9 किमी/घंटा

$$\text{नाव} = \frac{15 + 9}{2} = 12 \text{ किमी/घंटा} \quad \text{धारा} = \frac{15 - 9}{2} = 3 \text{ किमी/घंटा}$$

9. दौड़ (Races)

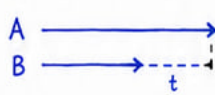
(i) A, B को x मीटर से हराता है

$$\frac{A}{B} = \frac{D}{D - x}$$



(ii) A, B को t सेकंड से हराता है

$$\frac{A}{B} = \frac{T}{T - t}$$



उदाहरण:

1 कमी की दौड़ में A, B को 200 मीटर से हराता है।

$$\frac{A}{B} = \frac{1000}{1000 - 200} = \frac{1000}{800} = 5 : 4$$

10. वृत्ताकार पथ (Circular Track)

• समान दिशा → $S_1 - S_2$

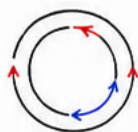
• विपरीत दिशा → $S_1 + S_2$

• प्रथम मिलन → परिधि

सापेक्ष चाल

• द्वितीय मिलन → $2 \times$ परिधि

सापेक्ष चाल



☆ अधिक अभ्यास करें, अधिक अंक प्राप्त करें!

समय, चाल एवं दूरी अध्याय के लगभग सभी प्रतियोगी परीक्षा प्रश्न इन्हीं सूत्रों पर आधारित होते हैं।

12. महत्वपूर्ण शॉर्टकट (Important Shortcuts)

यदि चाल x% बढ़ती है → समय में कमी = $\frac{x}{100 + x} \times 100\%$

यदि चाल x% घटती है → समय में वृद्धि = $\frac{x}{100 - x} \times 100\%$



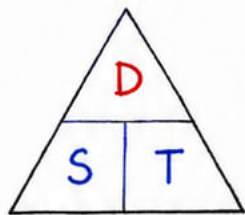
THE ONLY FORMULA SHEET YOU NEED

CORE FORMULA

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$\text{Distance} = \text{Speed} \times \text{Time}$$

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$



Remember:

- Cover the one you need
- Use the other two to find it.

Units: Speed $\rightarrow$ km/h or m/s
Distance $\rightarrow$ km or m Time $\rightarrow$ h or s

1. UNIT CONVERSION

$$1 \text{ km/hr} = \frac{5}{18} \text{ m/s}$$

$$1 \text{ m/s} = \frac{18}{5} \text{ km/hr}$$

2. RELATIVE SPEED

- Same direction $\rightarrow S_1 - S_2$
- Opposite direction $\rightarrow S_1 + S_2$

3. MEETING POINT CONCEPT

Case (i): Moving towards each other

$$A \rightarrow \quad \times \quad \leftarrow B$$

$$\text{Time} = \frac{D}{S_1 + S_2}$$

Case (ii): Moving in same direction

$$A \rightarrow \quad \rightarrow B$$

$$\text{Time} = \frac{D}{S_1 - S_2}$$

$$(S_1 > S_2)$$

Q) Two persons start from A & B, 100 km apart, speeds 40 & 60 km/hr towards each other.

$$\text{Time} = \frac{100}{40 + 60} = \frac{100}{100} = 1 \text{ hr}$$

4. TIME CONSTANT CONCEPT

If distance is constant:

$$S_1 T_1 = S_2 T_2$$

Q) Speed reduced in ratio 4:3
Time increases in ratio 3:4

5. SPEED CONSTANT CONCEPT

If speed is constant:

$$\frac{D_1}{D_2} = \frac{T_1}{T_2}$$

6. AVERAGE SPEED

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}}$$

! If distance 'a' and 'b' are covered with same speed,

$$\text{Average speed} = 2ab = \frac{2ab}{a+b}$$

11. BACKWARD & FORWARD

$$\text{Net speed} = \text{Forward} - \text{Backward}$$

Q) Boat goes 30 km downstream in 2 hrs, upstream in 3 hrs

$$\text{Down} = \frac{30}{2} = 15 \text{ km/hr} \quad \text{Boat} = 12.5 \text{ km/hr}$$

$$\text{Up} = \frac{30}{3} = 10 \text{ km/hr} \quad \text{Stream} = 2.5 \text{ km/hr}$$

7. PROBLEMS ON TRAINS

(i) Passing a pole/man



$$\text{Time} = \frac{\text{Length of train}}{\text{Speed}}$$

(ii) Passing a platform



$$\text{Time} = \frac{\text{Length of train} + \text{Length of platform}}{\text{Speed}}$$

(iii) Two trains opposite direction



$$\text{Time} = \frac{L_1 + L_2}{S_1 + S_2}$$

(iv) Same direction



$$\text{Time} = \frac{L_1 - L_2}{S_1 - S_2}$$

$(L_1 > L_2, S_1 > S_2)$

(v) Train 120 m crosses man in 8 sec

$$\text{Speed} = \frac{120}{8} = 15 \text{ m/s} = 54 \text{ km/hr}$$

8. BOATS & STREAMS

Speed of boat in still water = B

Speed of stream = S

Downstream speed = B + S

Upstream speed = B - S



$$\text{Boat speed} = \frac{D + U}{2}, \quad \text{Stream speed} = \frac{D - U}{2}$$

Q) Down = 15 km/hr, Up = 9 km/hr

$$\text{Boat} = (15 + 9)/2 = 24/2 = 12 \text{ km/hr}$$

$$\text{Stream} = (15 - 9)/2 = 6/2 = 3 \text{ km/hr}$$

9. RACES

Case (i): A beats B by x meters

$$\frac{A}{B} = \frac{D}{D - x}$$



Case (ii): A beats B by t sec

$$\frac{A}{B} = \frac{T}{T - t}$$

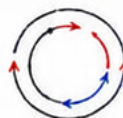


Q) A beats B by 200 m in 1 km race

$$\frac{A}{B} = \frac{1000}{1000 - 200} = \frac{1000}{800} = 5:4$$

10. CIRCULAR TRACK

- Same direction $\rightarrow (S_1 - S_2)$
- Opposite direction $\rightarrow (S_1 + S_2)$
- First meeting $\rightarrow$ relative speed
- Second meeting $\rightarrow 2 \times$ relative speed



12. IMPORTANT SHORTCUTS

If speed $\uparrow$ by x%, time $\downarrow$ by $\frac{x}{100+x} \%$

If speed $\downarrow$ by x%, time $\uparrow$ by $\frac{x}{100-x} \%$

Practice more, Score more!



1 मूल परिभाषाएँ

अनुपात = एक ही प्रकार की दो राशियों की तुलना

$$a : b = \frac{a}{b}$$

समानुपात :

$$a : b :: c : d \Rightarrow \frac{a}{b} = \frac{c}{d} \Rightarrow ad = bc$$



2 अनुपात के गुणधर्म

- $a : b = ka : kb$
- $a : b = \frac{a}{k} : \frac{b}{k}$
- $a : b = b : a$ (व्युत्क्रम)



3 संयुक्त अनुपात

$$\begin{aligned} a : b &= m : n \\ b : c &= p : q \\ \Rightarrow a : b : c &= mp : np : nq \end{aligned}$$



4 चकानुपात (संयुक्त अनुपात)

$$(a : b) \times (c : d) = ac : bd$$



5 किसी अनुपात में दी गई राशि को बाँटना

योग = S, अनुपात = a : b

$$\text{पहला भाग} = S \times \frac{a}{a+b}$$

$$\text{दूसरा भाग} = S \times \frac{b}{a+b}$$

(प्र.) 300 को 2 : 3 के अनुपात में बाँटें

$$\Rightarrow \text{कुल भाग} = 2 + 3 = 5$$

$$1 \text{ भाग} = \frac{300}{5} = 60$$

$$\text{भाग} = 2 \times 60 = 120, 3 \times 60 = 180$$



6 अंतर दिए जाने पर

(प्र.) $a : b = 3 : 5$, अंतर = 20

मान लें $a = 3x$, $b = 5x$

$$5x - 3x = 20$$

$$2x = 20$$

$$x = 10$$

$$a = 3 \times 10 = 30, b = 5 \times 10 = 50$$



7 गुणा (मल्टिपल) अनुपात

(प्र.) $a : b = 2 : 3$

$b : c = 4 : 5$

b (3 और 4 का LCM) = 12

$$a : b = (2 \times 4) : (3 \times 4) = 8 : 12$$

$$b : c = (4 \times 3) : (5 \times 3) = 12 : 15$$

$$\Rightarrow a : b : c = 8 : 12 : 15$$



8 योग दिए जाने पर (3 पदों में)

(प्र.) $a : b : c = 3 : 4 : 5$

योग = 240

$$\text{कुल भाग} = 3 + 4 + 5 = 12$$

$$1 \text{ भाग} = \frac{240}{12} = 20 \Rightarrow a = 3 \times 20 = 60$$

$$b = 4 \times 20 = 80$$

$$c = 5 \times 20 = 100$$



9 तुलना का प्रकार

(प्र.) यदि $a : b = 3 : 2$

तो $(4a + 5b) : (5a - 2b)$ ज्ञात करें

$a = 3$, $b = 2$ रखने पर

$$\Rightarrow (4 \times 3 + 5 \times 2) : (5 \times 3 - 2 \times 2)$$

$$\Rightarrow (12 + 10) : (15 - 4)$$

$$\Rightarrow 22 : 11 = 2 : 1$$



10 व्युत्क्रमानुपाती विभाजन

$$\text{यदि } x \propto y \Rightarrow \frac{x}{y} = \text{नियत}$$



(प्र.) यदि $x : y = 5 : 7$ तथा $x + y = 144$

$$\text{कुल भाग} = 5 + 7 = 12$$

$$1 \text{ भाग} = \frac{144}{12} = 12 \Rightarrow x = 5 \times 12 = 60, y = 7 \times 12 = 84$$

11 महत्वपूर्ण समताएँ (पहचानें)

यदि $a : b = m : n$

$$\bullet (a + b) : (a - b) = (m + n) : (m - n)$$

$$\bullet (a^2 - b^2) : (ab) = (m^2 - n^2) : (mn)$$



12 मिश्रित घात अनुपात

(प्र.) यदि $a : b = 2 : 3$

$$a^2 : b^2 = 2^2 : 3^2 = 4 : 9$$

$$\sqrt{a} : \sqrt{b} = \sqrt{2} : \sqrt{3}$$



13 परिवर्तन (Variation)

प्रत्यक्ष परिवर्तन

$$x \propto y \Rightarrow x = ky$$

व्युत्क्रम परिवर्तन

$$x \propto \frac{1}{y} \Rightarrow xy = k$$



14 महत्वपूर्ण शॉर्टकट

✓ जहाँ संभव हो, समान इकाइयों में बदलें

✓ जटिल प्रश्नों में उपस्थापन (Substitution) का प्रयोग करें

✓ यदि घात अलग-अलग हों, तो छोटी तुलना न करें

✓ LCM एक श्रृंखलाबद्ध (chain) अनुपात से समय बचाते हैं



15 सामान्य गलतियाँ

✗ इकाइयों की अनदेखी करना

✗ श्रृंखलाबद्ध (chain) अनुपात में गलत LCM लेना

✗ योग और अंतर के कहने में गलती करना

✗ विभाज्यता में इकाई स्थापत्य की गलती करना





1 BASIC PROPORTION

Proportion = Equality of two ratios

$$a : b = \frac{a}{b}$$

Therefore :

$$a : b :: c : d \Rightarrow \frac{a}{b} = \frac{c}{d} \Rightarrow ad = bc$$



2 PROPERTIES OF PROPORTION

- $a : b = ka : kb$
- $a : b = \frac{a}{k} : \frac{b}{k}$
- $a : b = b : a$ (Invertedo)



3 COMPOUND PROPORTION

$$a : b = m : n$$

$$b : c = p : q$$

$$\Rightarrow a : b : c = mp : np : nq$$



4 CONTINUED PROPORTION

$$(a : b) \times (c : d) = ac : bd$$



5 DIVIDING A GIVEN QUANTITY IN A GIVEN RATIO

Let ratio = S , Ratio = $a : b$

$$\text{First part} = S \times \frac{a}{a+b}$$

$$\text{Second part} = S \times \frac{b}{a+b}$$

(Q.) Divide 300 in the ratio 2 : 3

$$\Rightarrow \text{Total parts} = 2 + 3 = 5$$

$$1 \text{ part} = \frac{300}{5} = 60$$

$$\text{Parts} = 2 \times 60 = 120, 3 \times 60 = 180$$



6 INTERNAL DIVISION

(Q.) $a : b = 3 : 5$, Sum = 20

$$\text{Let } a = 3x, b = 5x$$

$$5x - 3x = 20$$

$$2x = 20$$

$$x = 10$$

$$a = 3 \times 10 = 30, b = 5 \times 10 = 50$$



7 MULTIPLE PROPORTION

(Q.) $a : b = 2 : 3$

$$b : c = 4 : 5$$

$$b \text{ (3 and 4 LCM)} = 12$$

$$a : b = (2 \times 4) : (3 \times 4) = 8 : 12$$

$$b : c = (4 \times 3) : (5 \times 3) = 12 : 15$$

$$\Rightarrow a : b : c = 8 : 12 : 15$$



8 DIRECT VARIATION (IN 3 TERMS)

(Q.) $a : b : c = 3 : 4 : 5$

Sum = 240

$$\text{Total parts} = 3 + 4 + 5 = 12$$

$$1 \text{ part} = \frac{240}{12} = 20 \Rightarrow a = 3 \times 20 = 60$$

$$b = 4 \times 20 = 80$$

$$c = 5 \times 20 = 100$$



9 DIVIDING INTO PARTS

(Q.) Divide $a : b = 3 : 2$

into $(4a + 5b) : (5a - 2b)$

Let $a = 3, b = 2$

$$\Rightarrow (4 \times 3 + 5 \times 2) : (5 \times 3 - 2 \times 2)$$

$$\Rightarrow (12 + 10) : (15 - 4)$$

$$\Rightarrow 22 : 11 = 2 : 1$$



10 INVERSE VARIATION

$$\text{If } x \propto y \Rightarrow \frac{x}{y} = \text{Constant}$$



(Q.) If $x : y = 5 : 7$ and $x + y = 144$

Total parts = $5 + 7 = 12$

$$1 \text{ part} = \frac{144}{12} = 12 \Rightarrow x = 5 \times 12 = 60, y = 7 \times 12 = 84$$

11 IMPORTANT FORMULAS (IDENTITIES)

If $a : b = m : n$

$$\bullet (a + b) : (a - b) = (m + n) : (m - n)$$

$$\bullet (a^2 - b^2) : (ab) = (m^2 - n^2) : (mn)$$



12 PROPORTION BETWEEN SQUARE AND CUBE

(Q.) If $a : b = 2 : 3$

$$a^2 : b^2 = 2^2 : 3^2 = 4 : 9$$

$$\sqrt{a} : \sqrt{b} = \sqrt{2} : \sqrt{3}$$



13 VARIATION (Variation)

$$\text{Direct variation } x \propto y \Rightarrow x = ky$$

$$\text{Inverse variation } x \propto \frac{1}{y} \Rightarrow xy = k$$



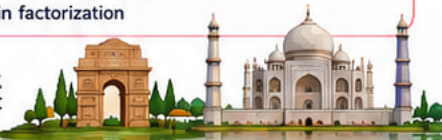
14 IMPORTANT SHORTCUTS

- ✓ When equal, compare and cancel
- ✓ Apply substitution in complex expressions
- ✓ If the ratio remains same, cancel it
- ✓ Use LCM or factorization in (chain) proportion



15 COMMON MISTAKES

- ✗ Canceling equal terms directly
- ✗ Not taking LCM in (chain) proportion
- ✗ Making mistakes in dividing into parts
- ✗ Not simplifying in factorization



RRB NTPC 30 Days Ultimate Study Plan

3D GEOMETRY & MENSURATION (VOLUME & SURFACE AREA) - FORMULA SHEET

1. CUBE & CUBOID

- ◆ **Cube:** Vol = a^3 | Total SA = $6a^2$ |
Lateral SA = $4a^2$ |
Diagonal = $a\sqrt{3}$
- ◆ **Cuboid:** Vol = LBH |
Total SA = $2(LB+BH+HL)$ |
Diagonal = $\sqrt{L^2+B^2+H^2}$
- ◆ **Area of 4 Walls:** $2(L+B) \times H$

3. CONE & FRUSTUM

- ◆ **Cone:** Vol = $\frac{1}{3} \pi r^2 h$ | CSA = $\pi r l$ |
Slant Height (l) = $\sqrt{r^2 + h^2}$
- ◆ **Frustum (Bucket):**
 ➡ Volume: $\frac{1}{3} \pi h (R^2 + r^2 + Rr)$
 ➡ CSA: $\pi l (R + r)$ where
 $l = \sqrt{h^2 + (R - r)^2}$

2. CYLINDER (Right Circular)

- ◆ **Volume:** $\pi r^2 h$
- ◆ **Curved Surface Area (CSA):**
 $2\pi r h$
- ◆ **Total Surface Area (TSA):**
 $2\pi r (h + r)$
- ◆ **Hollow Cylinder:**
Vol = $\pi h (R^2 - r^2)$ |
Total SA = $2\pi (R+r)(h + R-r)$

4. SPHERE & HEMISPHERE

- ◆ **Sphere:** Volume = $\frac{4}{3} \pi r^3$ |
Surface Area = $4\pi r^2$
- ◆ **Hemisphere:** Vol = $\frac{2}{3} \pi r^3$ |
CSA = $2\pi r^2$ | Total SA = $3\pi r^2$
- ◆ **Spherical Shell:**
Volume = $\frac{4}{3} \pi (R^3 - r^3)$

QUICK EXAM TIPS (3D):

- ✓ **Melting Concept:** When one solid is melted to form another, the VOLUME remains constant. (Volume 1 = Volume 2).
- ✓ **Volume Ratio:** If the ratio of radii of two spheres is $a:b$, then the ratio of their VOLUMES is $a^3:b^3$.
- ✓ **Euler's Formula:** For any solid: Faces + Vertices - Edges = 2 ($F + V - E = 2$).
- ✓ **Painting Cost:** Always calculate Total Surface Area (or CSA) for painting/costing questions, never Volume.

RRB NTPC 30 Days Ultimate Study Plan

2D GEOMETRY & MENSURATION - FORMULA SHEET



1. TRIANGLES (Advanced Properties)

- ◆ **General Area:** $\frac{1}{2} \times \text{Base} \times \text{Height}$
- ◆ **Equilateral Triangle:**
Area = $(\sqrt{3}/4)a^2$ | Height = $(\sqrt{3}/2)a$
- ✎ **Inradius (r):** $\frac{a}{2\sqrt{3}}$
- ✎ **Circumradius (R):** $\frac{a}{\sqrt{3}}$
- ◆ **Right-Angled Triangle:**
- ✎ **Circumradius (R):** Hypotenuse / 2
- ✎ **Inradius (r):** (Perpendicular + Base - Hypotenuse) / 2
- ◆ **Heron's Formula:**
 $\sqrt{s(s-a)(s-b)(s-c)}$ where $s = \frac{(a+b+c)}{2}$



2. QUADRILATERALS (4-Sided Shapes)

- ◆ **Square:** Area = a^2 (or $\frac{1}{2} \times d^2$) | Perimeter = $4a$ | Diagonal = $a\sqrt{2}$
- ◆ **Rectangle:**
Area = $L \times B$ | Perimeter = $2(L + B)$ | Diagonal = $\sqrt{L^2 + B^2}$
- ◆ **Rhombus:**
Area = $\frac{1}{2} \times d_1 \times d_2$ | Side = $\frac{1}{2} \times \sqrt{d_1^2 + d_2^2}$
- ✎ **Property:** $d_1^2 + d_2^2 = 4 \times \text{side}^2$
- ◆ **Trapezium:**
Area = $\frac{1}{2} \times (\text{Sum of Parallel Sides}) \times \text{Height}$
- ◆ **Parallelogram:**
Area = Base $\times$ Height (or $AB \sin \theta$)



3. CIRCLES & SECTORS

- ◆ **Area:** πr^2 | **Circumference:** $2\pi r$
- ◆ **Semi-Circle:**
Area = $\frac{1}{2} \pi r^2$ | Perimeter = $(\pi + 2)r$
- ◆ **Sector:**
Area = $\frac{\theta}{360} \times \pi r^2$ | Arc Length = $\frac{\theta}{360} \times 2\pi r$
- ◆ **Ring (Path):**
Area = $\pi(R^2 - r^2)$



4. POLYGONS (Multi-Sided)

- ◆ **Sum of Interior Angles:** $(n - 2) \times 180^\circ$
- ◆ **Each Interior Angle:** $\frac{[(n - 2) \times 180^\circ]}{n}$
- ◆ **Each Exterior Angle:** $360^\circ / n$
- ◆ **Number of Diagonals:** $n(n - 3) / 2$



QUICK EXAM TIPS (2D):

- ✓ **Area Ratio:** If the ratio of sides of two similar triangles is $a:b$, then the ratio of their AREAS is $a^2:b^2$.
- ✓ **Successive Increase:** If length increases by $x\%$ and breadth by $y\%$, then Area increases by $[x + y + (xy/100)]\%$.
- ✓ **Incircle/Circumcircle:** For any triangle, Area = $r \times s$ (where r = inradius, s = semi-perimeter).
- ✓ **Polygon Tip:** For any regular polygon, Sum of Exterior Angles is ALWAYS 360° .

STATISTICS: EMPIRICAL & GROUPED DATA FORMULAS (WITH VARIABLE DEFINITIONS)

EMPIRICAL RELATION

$$\text{Mode} = 3(\text{Median}) - 2(\text{Mean})$$

VARIABLES DEFINED

Mode = Most frequent value; **Median** = Middle value; **Mean** = Average value

GROUPED MEAN

$$\bar{x} = \frac{\sum fx}{\sum f}$$

VARIABLES

$\bar{x}$ = Mean; Σ = Summation;
 f = Frequency;
 x = Class Mark (Midpoint);
 Σf = Total frequency (N)

$$\bar{x} = A + \frac{\sum fd}{\sum f} \cdot h$$

A = Assumed Mean;
 $d = (x - A) / h$;
 h = Class size

GROUPED MEDIAN

$$\text{Median} = l + \left[\frac{\frac{N}{2} - cf}{f} \right] \cdot h$$

VARIABLES

l = Lower limit of median class;
 $N = \Sigma f$ (Total frequency);
 cf = Cumulative frequency of preceding class;
 f = Frequency of median class;
 h = Class size

GROUPED MODE

$$\text{Mode} = l + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \cdot h$$

VARIABLES

l = Lower limit of modal class;
 f_1 = Frequency of modal class;
 f_0 = Frequency of preceding class;
 f_2 = Frequency of succeeding class;
 h = Class size

Elementary Statistics: Question types and formulas - Important for SSC and Railway Exams

ELEMENTARY STATISTICS: KEY CONCEPTS & TRICKS

DATA ANALYSIS TYPES

MEAN (AVERAGE)

Q: Find mean of 5, 8, 12, 15, 15, 20.

FORMULA: Mean $(\bar{x}) = \frac{\sum x}{N}$

SHORT TRICK: Assume a mean (e.g., 12), calculate deviations, average deviations, add to assumed mean.

MEDIAN (MIDDLE VALUE)

Q: Find median of 3, 7, 12, 12, 15, 18.

FORMULA:
For odd N: $((N+1)/2)$ th.
For even N: $N/2$ = Average of $(N/2)$ th and $((N/2)+1)$ th terms.

SHORT TRICK: Arrange in ascending order. Find the middle term(s) directly.

MEASURES OF CENTRAL TENDENCY & DISPERSION

MODE (MOST FREQUENT)

Q: Find mode of 2, 5, 8, 5, 9, 5, 2.

FORMULA: Value with highest frequency.

SHORT TRICK: Simply count the frequency of each number; the highest count is the mode.

RANGE (SPREAD)

Q: Find range of 10, 25, 5, 30, 15.

FORMULA: Range = Maximum Value - Minimum Value.

SHORT TRICK: Quickly check largest and smallest numbers and subtract.

STANDARD DEVIATION (SD)

Q: Find SD of 2, 4, 6, 8, 10.

FORMULA: $\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{N}}$

SHORT TRICK: Use assume mean method or calculate mean, deviations, average them, take square root.

VARIANCE

Q: Find variance if SD is 5.

FORMULA: σ^2 .

SHORT TRICK: Square the Standard Deviation value directly.

RRB NTPC 30 Days Ultimate Study Plan

COMPLETE PROFIT, LOSS & DISCOUNT FORMULA SHEET

1. BASIC TERMINOLOGY

- ◆ **Cost Price (CP):** The price at which an article is bought.
- ◆ **Selling Price (SP):** The price at which an article is sold.
- ◆ **Marked Price (MP):** The price printed on the label (List Price).

2. BASIC FORMULAS

- ◆ **Profit = SP - CP** (When $SP > CP$).
 - ◆ **Loss = CP - SP** (When $CP > SP$).
 - ◆ **Profit % = (Profit / CP) × 100**
 - ◆ **Loss % = (Loss / CP) × 100**
- ⚠ Note: Profit and Loss % are always calculated on CP unless stated otherwise.

3. DIRECT SP & CP CALCULATION

- ◆ **SP = CP × [(100 + Profit%) / 100]**
- ◆ **SP = CP × [(100 - Loss%) / 100]**
- ◆ **CP = [100 / (100 + Profit%)] × SP**
- ◆ **CP = [100 / (100 - Loss%)] × SP**

4. DISCOUNT & MARKED PRICE

- ◆ **Discount = MP - SP**
- ◆ **Discount % = (Discount / MP) × 100**
- ◆ **SP = MP × [(100 - Discount%) / 100]**

5. THE "GOLDEN RELATION" (MP & CP)

- ◆ **CP / MP = (100 - Discount%) / (100 + Profit%).**
(This formula is the most used in SSC CGL/CHSL exams!)

6. SUCCESSIVE DISCOUNTS

- ◆ If two discounts are $x\%$ and $y\%$:
Single Equivalent Discount = $[x + y - (xy/100)] \%$.

7. DISHONEST DEALER SHORTCUT

- ◆ **Profit % = [(True Value - False Value) / False Value] × 100**
(Example: Using 900g instead of 1kg)

QUICK EXAM TRICKS:

✓ **No Profit, No Loss:** When two items are sold at the same price (SP), one at $x\%$ profit and another at $x\%$ loss, there is **ALWAYS** a loss.

👉 Total Loss % = $(x / 10) \%$.

✓ **Buy X, Get Y Free:**

👉 Discount % = $[\text{Free Articles} / \text{Total Articles (X+Y)}] \times 100$.

RRB NTPC 30 Days Ultimate Study Plan

🔥 साधारण ब्याज (SI) और चक्रवृद्धि ब्याज (CI) - कम्पलीट शीट 🔥

1. साधारण ब्याज (SIMPLE INTEREST - SI)

- ◆ $SI = \frac{\text{मूलधन} \times \text{दर} \times \text{समय}}{100}$
 👉 $SI = (P \times R \times T) / 100$
- ◆ मिश्रधन (Amount) = मूलधन (P) + ब्याज (SI)
- ◆ दर (Rate) = $\frac{\text{ब्याज} \times 100}{\text{मूलधन} \times \text{समय}}$
- ◆ समय (Time) = $\frac{\text{ब्याज} \times 100}{\text{मूलधन} \times \text{दर}}$

2. चक्रवृद्धि ब्याज (COMPOUND INTEREST - CI)

- ◆ मिश्रधन (A) = $P \left[1 + \frac{R}{100} \right]^T$
- ◆ CI = मिश्रधन - मूलधन
- ◆ $CI = P \left[\left(1 + \frac{R}{100} \right)^T - 1 \right]$

3. छमाही और तिमाही गणना (Special Cases)

- ◆ छमाही (Half-Yearly): दर आधी ($R/2$) और समय दुगुना ($2T$) हो जाता है।
- ◆ तिमाही (Quarterly): दर चौथाई ($R/4$) और समय चार गुना ($4T$) हो जाता है।

4. अंतर वाले शॉर्टकट्स (EXAM FAVOURITE)

परीक्षा में अक्सर CI और SI के बीच का अंतर पूछा जाता है:

- ◆ 2 वर्ष के लिए अंतर = $P (R/100)^2$
- ◆ 3 वर्ष के लिए अंतर = $[P (R/100)^2] \times [(300 + R) / 100]$

5. किस्त (INSTALLMENT) फॉर्मूला

- ◆ साधारण ब्याज की किस्त:
 वार्षिक भुगतान = $\frac{100 \times \text{मिश्रधन}}{100T + RT(T-1)/2}$
- ◆ चक्रवृद्धि ब्याज की किस्त:
 $P = \left[\frac{x}{(1 + R/100)} \right] + \left[\frac{x}{(1 + R/100)^2} \right] \dots$
 (जहाँ x प्रत्येक किस्त का मान है)।

6. धन का गुना होना (Shortcuts)

- ◆ साधारण ब्याज: यदि धन T वर्षों में 'n' गुना हो जाता है:
 👉 दर = $[(n - 1) / T] \times 100$
- ◆ चक्रवृद्धि ब्याज: यदि धन T वर्षों में 'n' गुना होता है, तो n^m गुना होने में ($m \times T$) वर्ष लगेंगे।

QUICK EXAM TRICKS:

- ✓ Effective Rate for 2 years (CI): $[x + y + (xy/100)]\%$
 (Example: 10% for 2 years CI is $10 + 10 + 1 = 21\%$)
- ✓ Tree Method: Always use the "Tree Method" or "Ratio Method" for CI to avoid long power calculations.
- ✓ CI Ratios:
 - 👉 For 2 years: 2 : 1
 - 👉 For 3 years: 3 : 3 : 1
 - 👉 For 4 years: 4 : 6 : 4 : 1

RRB NTPC 30 Days Ultimate Study Plan

INTEREST FORMULA SHEET (SI & CI)

1. SIMPLE INTEREST (SI)

- ◆ $SI = \frac{P \times R \times T}{100}$
- ◆ **Amount (A)** = $P + SI$
- ◆ **Rate (R)** = $\frac{SI \times 100}{P \times T}$
- ◆ **Time (T)** = $\frac{SI \times 100}{P \times R}$

(Where P = Principal, R = Rate %, T = Time in years)

2. COMPOUND INTEREST (CI)

- ◆ **Amount (A)** = $P \left[1 + \frac{R}{100} \right]^T$
- ◆ **CI** = Amount - Principal
- ◆ **CI** = $P \left[\left(1 + \frac{R}{100} \right)^T - 1 \right]$

3. HALF-YEARLY & QUARTERLY (CI)

- ◆ **Half-Yearly:**
Rate = $R/2$, Time = $2T$
- ◆ **Quarterly:**
Rate = $R/4$, Time = $4T$

4. THE "DIFFERENCE" SHORTCUTS (Very Important)

Most exam questions ask for the difference between CI and SI:

- ◆ **For 2 Years:** Difference = $P (R/100)^2$
- ◆ **For 3 Years:** Difference = $[P (R/100)^2] \times [(300 + R) / 100]$

5. INSTALLMENT FORMULAS

- ◆ **Simple Interest Installment:**
Annual Payment = $\frac{100 \times \text{Amount}}{100T + RT(T-1)/2}$
- ◆ **Compound Interest Installment:**
 $P = \left[\frac{x}{(1 + R/100)} \right] + \left[\frac{x}{(1 + R/100)^2} \right] \dots$
(Where x is the value of each installment)

6. TIMES OF MONEY (Shortcuts)

- ◆ **Simple Interest:** If money becomes 'n' times in T years:
Rate = $[(n - 1) / T] \times 100$
- ◆ **Compound Interest:** If money becomes 'n' times in T years, it will become n^m times in $(m \times T)$ years.

QUICK EXAM TRICKS:

- ✓ **Effective Rate for 2 years (CI):** $[x + y + (xy/100)]\%$
(Example: 10% for 2 years CI is $10 + 10 + 1 = 21\%$)
- ✓ **Tree Method:** Always use the "Tree Method" or "Ratio Method" for CI to avoid long power calculations.
- ✓ **CI Ratios:**
 - 👉 For 2 years: 2 : 1
 - 👉 For 3 years: 3 : 3 : 1
 - 👉 For 4 years: 4 : 6 : 4 : 1

RRB NTPC 30 Days Ultimate Study Plan

COMPLETE AVERAGE FORMULA SHEET

1. BASIC CONCEPT

- ◆ $\text{Average} = \frac{\text{Sum of all observations}}{\text{Total number of observations}}$
- ◆ $\text{Total Sum} = \text{Average} \times \text{Number of terms}$

2. NUMBER SYSTEM SHORTCUTS

- ◆ Avg of first 'n' natural numbers
 $= \frac{(n+1)}{2}$
- ◆ Avg of first 'n' even numbers $= (n+1)$
- ◆ Avg of first 'n' odd numbers $= n$
- ◆ Avg of first 'n' natural square numbers
 $= \frac{(n+1)(2n+1)}{6}$
- ◆ Avg of first 'n' natural cube numbers
 $= \frac{n(n+1)^2}{4}$

3. CONSECUTIVE SERIES (Same Difference)

If the difference between terms is constant (AP series):

- ◆ $\text{Average} = (\text{First Term} + \text{Last Term}) / 2$
(Works for consecutive numbers, even/odd series, or multiples of any number)

4. AVERAGE SPEED

- ◆ If distances are equal (2 speeds):
 $\text{Avg Speed} = \frac{2xy}{x+y}$
- ◆ If distances are equal (3 speeds):
 $\text{Avg Speed} = \frac{3xyz}{xy+yz+zx}$
- ◆ General Formula: $\frac{\text{Total Distance}}{\text{Total Time}}$

5. JOINING / LEAVING (Shortcut)

- ◆ New Person Joins:
 $\text{New Value} = \text{New Avg} + (\text{Initial Number of people} \times \text{Change in Avg})$
- ◆ Person Leaves:
 $\text{Left Person's Value} = \text{Old Avg} + (\text{Remaining Number of people} \times \text{Decrease in Avg})$

6. CRICKET SPECIAL

- ◆ $\text{Batting Average} = \frac{\text{Total Runs}}{\text{Total Innings (excluding Not Outs)}}$
- ◆ $\text{Bowling Average} = \frac{\text{Total Runs conceded}}{\text{Total Wickets taken}}$

QUICK REVISION TIPS:

- ✓ If every number in a set is increased by 'k', the Average also increases by 'k'.
- ✓ If every number is multiplied by 'k', the Average is also multiplied by 'k'.
- ✓ **Replacement Trick:** If one person is replaced by another:
 $\text{New Person} = \text{Weight of Removed Person} + (\text{Total Members} \times \text{Increase in Avg})$

RRB NTPC 30 Days Ultimate Study Plan

COMPLETE AVERAGE FORMULA SHEET

1. बेसिक फॉर्मूला

- ◆ $\text{Average} = \frac{\text{पदों का कुल योग}}{\text{पदों की कुल संख्या}}$
- ◆ $\text{Sum (योग)} = \text{Average} \times \text{Number of terms}$

2. संख्या पद्धति पर आधारित

- ◆ प्रथम 'n' प्राकृतिक संख्याओं का औसत = $(n + 1) / 2$
- ◆ प्रथम 'n' सम (Even) संख्याओं का औसत = $(n + 1)$
- ◆ प्रथम 'n' विषम (Odd) संख्याओं का औसत = n
- ◆ प्रथम 'n' प्राकृतिक संख्याओं के वर्गों का औसत = $\frac{(n + 1)(2n + 1)}{6}$
- ◆ प्रथम 'n' प्राकृतिक संख्याओं के घनों (Cubes) का औसत = $\frac{n(n + 1)^2}{4}$

3. लगातार संख्याओं का औसत (Consecutive Numbers)

अगर संख्याओं के बीच का अंतर समान हो (जैसे: 2, 4, 6, 8...):

- ◆ $\text{Average} = (\text{प्रथम पद} + \text{अंतिम पद}) / 2$

4. औसत चाल

- ◆ जब दूरियाँ समान हों:
 $\text{Avg Speed} = \frac{2xy}{x + y}$
(जहाँ x और y दो अलग-अलग चालें हैं)
- ◆ जब 3 समान दूरियाँ हों:
 $\text{Avg Speed} = \frac{3xyz}{xy + yz + zx}$

5. ग्रुप में शामिल होना/छोड़ना

- ◆ नया सदस्य आने पर:
नया मान = नया औसत + (संख्या में वृद्धि × औसत में वृद्धि)
- ◆ पुराना सदस्य जाने पर:
गए व्यक्ति का मान = पुराना औसत + (पुरानी संख्या × औसत में कमी)

6. बल्लेबाज और गेंदबाज

- ◆ बल्लेबाजी औसत:
कुल रन / कुल आउट (पारी)
- ◆ गेंदबाजी औसत:
कुल दिए गए रन / कुल विकेट

QUICK REVISION TRICKS:

- ✓ यदि 'n' संख्याओं का औसत 'A' है और हर संख्या में 'K' जोड़ दिया जाए, तो नया औसत = $(A + K)$ होगा।
- ✓ यदि हर संख्या को 'K' से गुणा कर दें, तो नया औसत = $(A \times K)$ होगा।
- ✓ **Concept of Deviation:** बड़े कैलकुलेशन से बचने के लिए एक काल्पनिक औसत (Assumed Mean) मानकर विचलन (Deviation) निकालें।

RRB NTPC 30 Days Ultimate Study Plan

ALGEBRA IMPORTANT FORMULAS

1. BASIC IDENTITIES (SQUARES)

$$(a + b)^2 = a^2 + b^2 + 2ab$$

$$(a - b)^2 = a^2 + b^2 - 2ab$$

$$a^2 - b^2 = (a + b)(a - b)$$

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

$$(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$$

$$(a + b)^2 - (a - b)^2 = 4ab$$

2. CUBIC IDENTITIES

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$(a + b)^3 = a^3 + b^3 + 3ab(a + b)$$

$$(a - b)^3 = a^3 - b^3 - 3ab(a - b)$$

3. THE "KING" FORMULA (SSC Special)

If $[x - \frac{1}{x} = k]$, then:

$$\gg x^2 + \frac{1}{x^2} = k^2 + 2$$

$$\gg x^3 - \frac{1}{x^3} = k^3 + 3k$$

$$\gg x^4 + \frac{1}{x^4} = (k^2 + 2)^2 - 2$$

4. RECIPROCAL SHORTCUTS ($x + 1/x$)

If $[x + \frac{1}{x} = k]$, then:

$$\gg x^2 + \frac{1}{x^2} = k^2 - 2$$

$$\gg x^3 + \frac{1}{x^3} = k^3 - 3k$$

$$\gg x^4 + \frac{1}{x^4} = (k^2 - 2)^2 - 2$$

$$\gg x^5 + \frac{1}{x^5} = \left(x^2 + \frac{1}{x^2}\right)\left(x^3 + \frac{1}{x^3}\right) - \left(x + \frac{1}{x}\right)$$

5. QUADRATIC EQUATIONS (2 ROOTS)

For equation: $ax^2 + bx + c = 0$

◆ Sum of roots ($\alpha + \beta$) = $-\frac{b}{a}$

◆ Product of roots ($\alpha\beta$) = $\frac{c}{a}$

◆ Discriminant (D) = $b^2 - 4ac$

6. CUBIC EQUATIONS (3 ROOTS)

For equation: $ax^3 + bx^2 + cx + d = 0$

1 Sum ($\alpha + \beta + \gamma$) = $-\frac{b}{a}$

2 Sum of Products ($\alpha\beta + \beta\gamma + \gamma\alpha$) = $\frac{c}{a}$

3 Product ($\alpha\beta\gamma$) = $-\frac{d}{a}$

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

★ Golden Condition: If $(a + b + c) = 0$, then:

👉 $a^3 + b^3 + c^3 = 3abc$

CALENDAR

- Every century is not leap year.

Ex:- 100 years $\rightarrow$ 5 odd day

200 $\rightarrow$ 3 odd day

300 $\rightarrow$ 1 odd day

- Every 4th century is leap year

Ex:- 400, 800, 1200, 1600, 2000 $\dots \Rightarrow$ 0 odd day.

- Up to 1900 $\Rightarrow$ 1 odd day

- 1 normal year $\Rightarrow$ 1 odd day • 1 leap year $\Rightarrow$ 2 odd d.

month odd day

Jan	$\rightarrow$ 3
Feb	$\rightarrow$ 0 normal, 1 Ly
Mar	$\rightarrow$ 3
Apr	$\rightarrow$ 2
May	$\rightarrow$ 3
June	$\rightarrow$ 2
July	$\rightarrow$ 3
Aug	$\rightarrow$ 3
Sep	$\rightarrow$ 2
Oct	$\rightarrow$ 3
Nov	$\rightarrow$ 2
Dec	$\rightarrow$ 3

Week odd day

Sun	$\rightarrow$ 0
Mon	$\rightarrow$ 1
Tues	$\rightarrow$ 2
Wed	$\rightarrow$ 3
Thur	$\rightarrow$ 4
Fri	$\rightarrow$ 5
Sat	$\rightarrow$ 6

(Q.) 2 Oct 1869. days?

	odd days
1600	$\rightarrow$ 0
200	$\rightarrow$ 3
for 68 LY	$\rightarrow 17 \times 2 = 34$
years Not. year	$\rightarrow$ 51
month up to Sep	$\rightarrow$ 21
days	$\rightarrow$ 2

$$7 \overline{) 111 (15} \\ \underline{105} \\ 6 \rightarrow \text{Sat.}$$

CLOCK

- Hrs hand in 1 hr $\rightarrow 30^\circ$ Angle
1 min $\rightarrow \frac{1}{2}^\circ$ Angle

- Min hand in 5 min $\rightarrow 30^\circ$ Angle
1 min $\rightarrow 6^\circ$ Angle

coincide = $5x \times \frac{12}{11}$

Right angle = $(5x \pm 15) \times \frac{12}{11}$
+ve $\rightarrow$ 1st ans
-ve $\rightarrow$ 2nd ans

opposite = $(5x \pm 30) \times \frac{12}{11}$

If $x > 6 \rightarrow$ Take -ve sign
 $x < 6 \rightarrow$ " +ve "

Mirror image = 12 - given

* General formula

$$\left[5x \pm \left(\frac{D}{6} \right) \right] \times \frac{12}{11}$$

Put, $D \rightarrow 0^\circ$ (coincide)

$D \rightarrow 90^\circ$ (Right Ls)

$D \rightarrow 180^\circ$ (opposite)

(Q.) coincide, right Ls, opposite b/n 4'o clock & 5'o clock.

$\Rightarrow$ Put $x = 4$ then solve.

for Right angle

$$(5x + 15) \times \frac{12}{11} = \frac{420}{11} = 38 \frac{2}{11} \Rightarrow 4:38 \frac{2}{11} \text{ Ans}$$

$$(5x - 15) \times \frac{12}{11} = \frac{60}{11} = 11 \frac{60}{55} = 5 \frac{5}{11} \Rightarrow 4:5 \frac{5}{11} \text{ Ans}$$

BLOOD RELATION

A^+ $\rightarrow$ for male

A^- $\rightarrow$ for female



Don't Judge gender by name

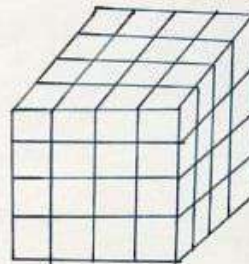
Bhanja, Bhateja $\Rightarrow$ Nephew

Bhanji, Bhatiji $\Rightarrow$ Niece

Relation from mother side $\Rightarrow$ Maternal

Relation from father side $\Rightarrow$ Paternal

CUBE



64 equal parts

$$\sqrt[3]{64} = 4 \Rightarrow \text{It means one side has 4 cube.}$$

• Total Blocks = $l \times b \times h$

- 3 side painted = 8
- 2 side painted = $4[(1-2) + (b-2) + (h-2)]$
- 1 side " = $2[(1-2)(b-2) + (b-2)(h-2) + (h-2)(1-2)]$
- 0 side painted = $(1-2)(b-2)(h-2)$

DICE

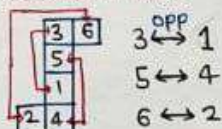
- When one no. is common on both the dices.



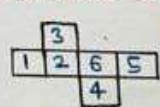
$$\Rightarrow \begin{pmatrix} 4 & 2 \\ 5 & 3 \end{pmatrix} \text{ opp. opp.}$$

Open dice

(i) General dice



(ii) standard dice:- opposite



surface sum should be 7.

CHESS BOARD

$n \times n$ board

- No. of Squares = Σn^2
- No. of Rectangles = Σn^3
- No. of types of rectangles = Σn

SYLLOGISM

+ve statement:- (i) All (ii) Some

-ve statement:- (i) No (ii) Some not

Some statement are:- Replacement of some:-

many, most of, at least, generally more, few

Replacement of word All:-

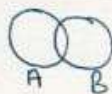
each, every, 100%.

- (i) All A is B
- (ii) Some A is B
- (iii) No A is B
- (iv) Some A is not B
- (v) only A is B
- (vi) only a few A is B

(1) All A is B:- (2) Some A is B:-



concⁿ:- 1. Some B are A
2. Some A are B

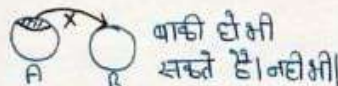


concⁿ:- (i) Some B are A

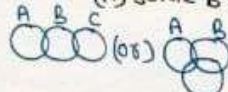
(3) No A is B:- (4) Some A is not B:-



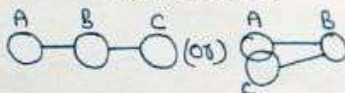
concⁿ:- (i) No B is A
(ii) Some A are not B
(iii) Some B are not A



Ex:- stat:- (i) Some A is B
(ii) Some B is C



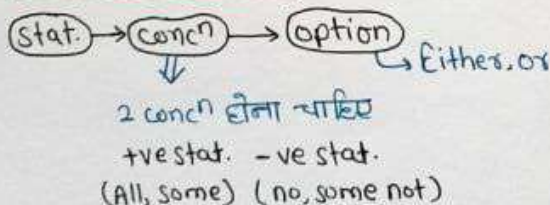
Ex:- stat:- (i) No A is B
(ii) No B is C



concⁿ:- (i) Some C are B [100% True]
(ii) No A is B [100% false]
(iii) No A is C [can't say]

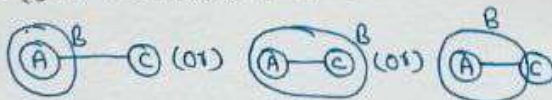
concⁿ:- (i) No B are A [100% sure]
(ii) Some C are not B [100% sure]
(iii) No A is C [CNS]

Either, or concept:-



दोनों stat. CNS वाले concn होना चाहिए।
element common [मूर्दे समान होना चाहिए]

Ex:- stat:- (i) All A is B (ii) No A is C



concⁿ:- (i) No B is C [CNS]
(ii) Some B are C [CNS]

(A) If only concn I follows (B) only concn II follows

(C) If either I (or) II follows

⇒ point (i) 2 concn → satisfied

(ii) one +ve & -ve → satisfied

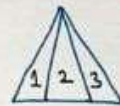
(iii) both stat. CNS → satisfied

(iv) common element → satisfied

COUNTING OF FIGURE

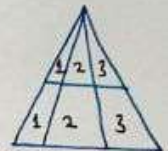
Triangle

Type(i)



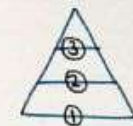
⇒ 1+2+3 ⇒ 6 Ans

Type-(iii)



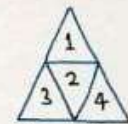
⇒ 12 Ans

Type(ii)

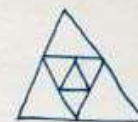


⇒ 3 Ans

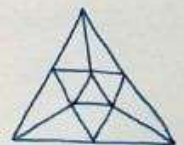
Type(iv)



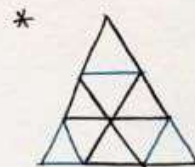
⇒ ऊपर-4
⇒ बाएर-1
⇒ 5 Ans



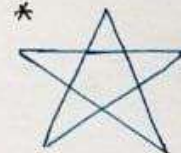
⇒ ऊपर-4
⇒ " -4
⇒ बाएर-1
⇒ 9 Ans



⇒ 4+4+1+6



⇒ 10+3=13 Ans

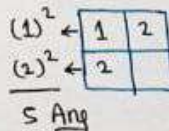


⇒ 5x2=10 Ans
vertex

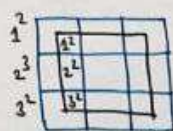
एक vertex से 2 triangle बन रहे हैं।

Square

Type-1 [sym fig] Row = column



5 Ans

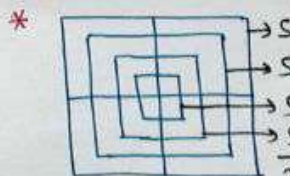


⇒ 28 Ans

Type-2 [Non-sym fig] Row ≠ column



3x4=12
2x3=6
1x2=2
20 Ans



20 Ans

HALL OF FAME !

MEET THE ACHIEVERS

FROM SSC TO RAILWAYS, BANKING TO STATE EXAMS

OUR TOPPERS SHINE EVERYWHERE !

 ROHIT CHADHAR AIR 1 SSC CHSL 2024	 MANISH KUMAR AIR 1 SSC CPO 2024	 ASHISH TIWARI AIR 2 SSC CGL 2024	 NIRANJAN JAIN AIR 1 SBI PO 2024	 MOHAN KUMAR AIR 1 SSC JE 2024 (ME)
 KANHAIYA KUMAR AIR 1 BSPHCL JAC	 LAXMI PRIYA AIR 1 OSSC JE 2024	 VIVEK SINGH LODHI AIR 8 SSC STENO 2024	 VIVEK SINGH AIR 7 MP POLICE CONSTABLE 2024	 PRIYA AIR 19 OSSSB JJA 2024
 SACHIN BAINSLA AIR 55 UPSC CDS II 2024	 KOMAL MUDGAL IBPS PO 2024	 GAUTAM CHHABRA SBI CLERK 2024	 AAKASH BIST RRB JE 2024	 VIKAS SINGH RRB TECHNICIAN GRADE - I
 RANJAN KUMAR BIHAR ASI STENO 2024	 CHIRAG SHARMA NIACL ASSISTANT 2024	 ARSHA SAXENA ALLAHABAD HC GROUP D	 SWAGAT NAYAK OSSC CGL 2024	 SOORAJ SINGH UP POLICE CONSTABLE 2023

THEY DID IT. YOU CAN TOO

READ THEIR STORIES !

